

Norma $\mathcal{L}^p$

Definición 1 (Norma $\mathcal{L}^p$). Sea (X, Σ, μ) un espacio de medida, $p \in [1, \infty]$ y $f: X \rightarrow \mathbb{R}$ medible, $\|f\|_p$ es la norma $\mathcal{L}^p$ de f

$$\Leftrightarrow \begin{cases} \|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p} & \text{si } p \in [1, \infty) \\ \|f\|_\infty = \text{ess sup } |f| = \inf_{\substack{A \in \Sigma \\ \mu(A)=0}} \left\{ \sup_{x \in A^c} |f(x)| \right\} & \text{si } p = \infty. \end{cases}$$